3. DRYING OF SOLIDS BY THERMAL VAPORISATION

Drying is the removal of moisture (either water or other volatile compounds) from solids, solutions, slurries and pastes. The drying of solids is usually the final step in the processing of solid materials. As much liquid as possible is removed mechanically (this is the cheapest method) before the material is fed to a *heated* drier.

3.1 Types of Driers and wet solids

A multitude of *different types* of driers is available depending on the following aspects:

1. Different forms of solids - granules, powders, crystals, slabs, continuous sheets
2. The liquid to be evaporated may be on the surface (*e.g.* with crystals) or inside the solid (*e.g.* with polymer sheets), or both
3. The final dry product might be delicate and possibly unstable in a hot environment (e.g. bioproducts).

Two main categories of wet solids can be distinguished:

i. Solids that hold moisture in open pores between particles: during drying the solid is unaffected by moisture removal.

ii. Solids that dissolve moisture, trap moisture in fibers or very fine pores: these materials are affected by moisture removal, often shrinking when dried and swelling when wetted.

Classification of types of driers is difficult; some may operate:

- Continuously, or batch-wise,
- with or without the agitation of the solids being dried,
- with or without operation under vacuum to reduce drying temperatures.

However, a major distinction can be made between:

1. **Adiabatic** (or *direct*) driers.

The wet solids are exposed directly to a hot gas, which can be blown:

- *across* the surface of the solids - *cross circulation* drying, **or**
- *through* the bed of solids supported on a screen - *through circulation* drying. The bed might be a moving one, *e.g.*:

  - downward showering of particles through a rising gas stream

  - fluidised bed drier (particles remain in the bed)

  - entrainment of particles upwards in a high velocity gas stream.

2. **Non-adiabatic** (or *indirect*) driers

The only gas involved is the vaporised liquid and heat is transferred to the solids indirectly *via*:

- a heated metal surface in contact with the solids (the surface is heated by condensing steam or electrical coils etc.), or
- radiant or micro-wave energy.

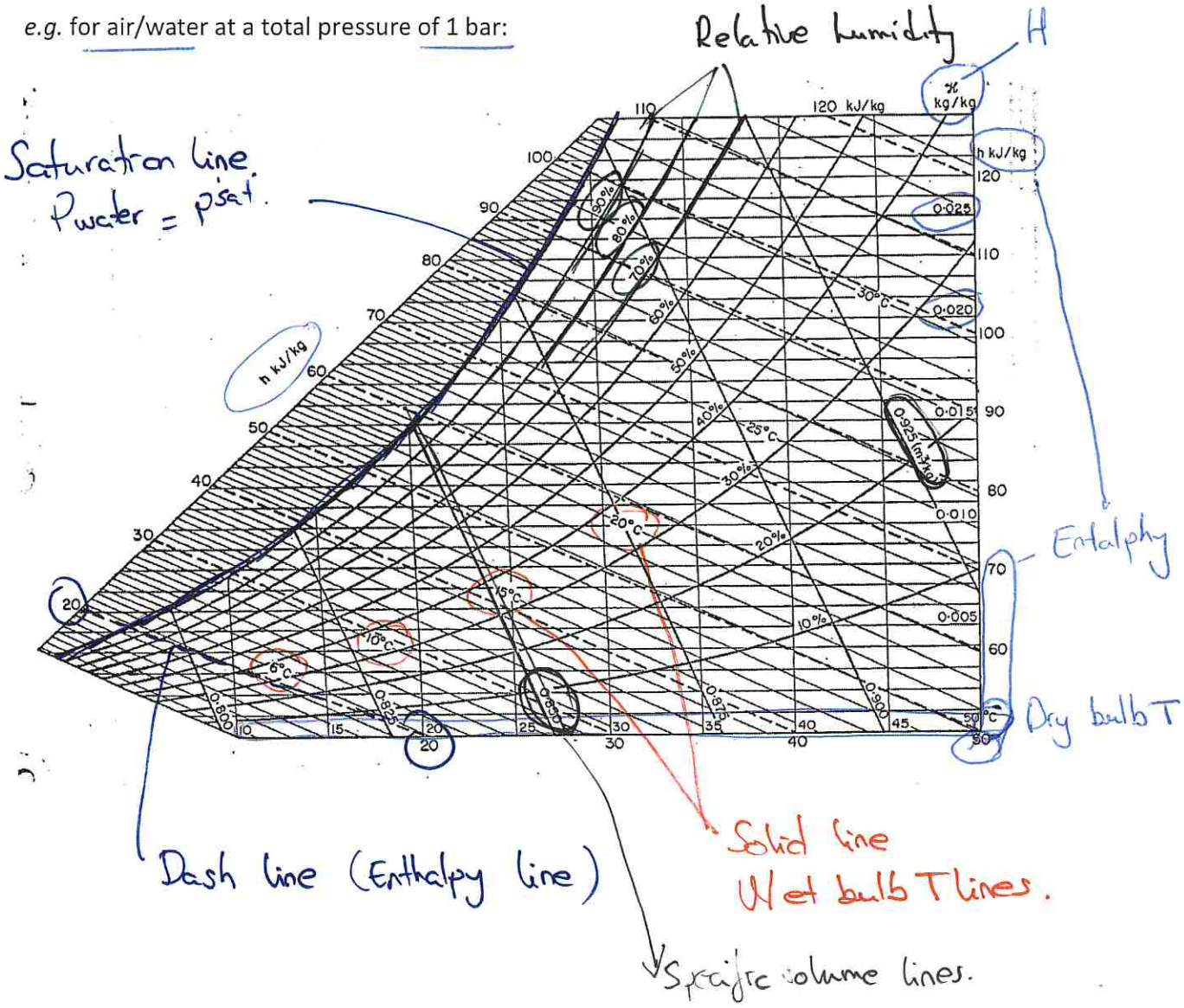
The solids are either spread on a heated surface, or moved over the surface by an agitator, or a screw or paddle conveyor, or by motion in a rotary drier.

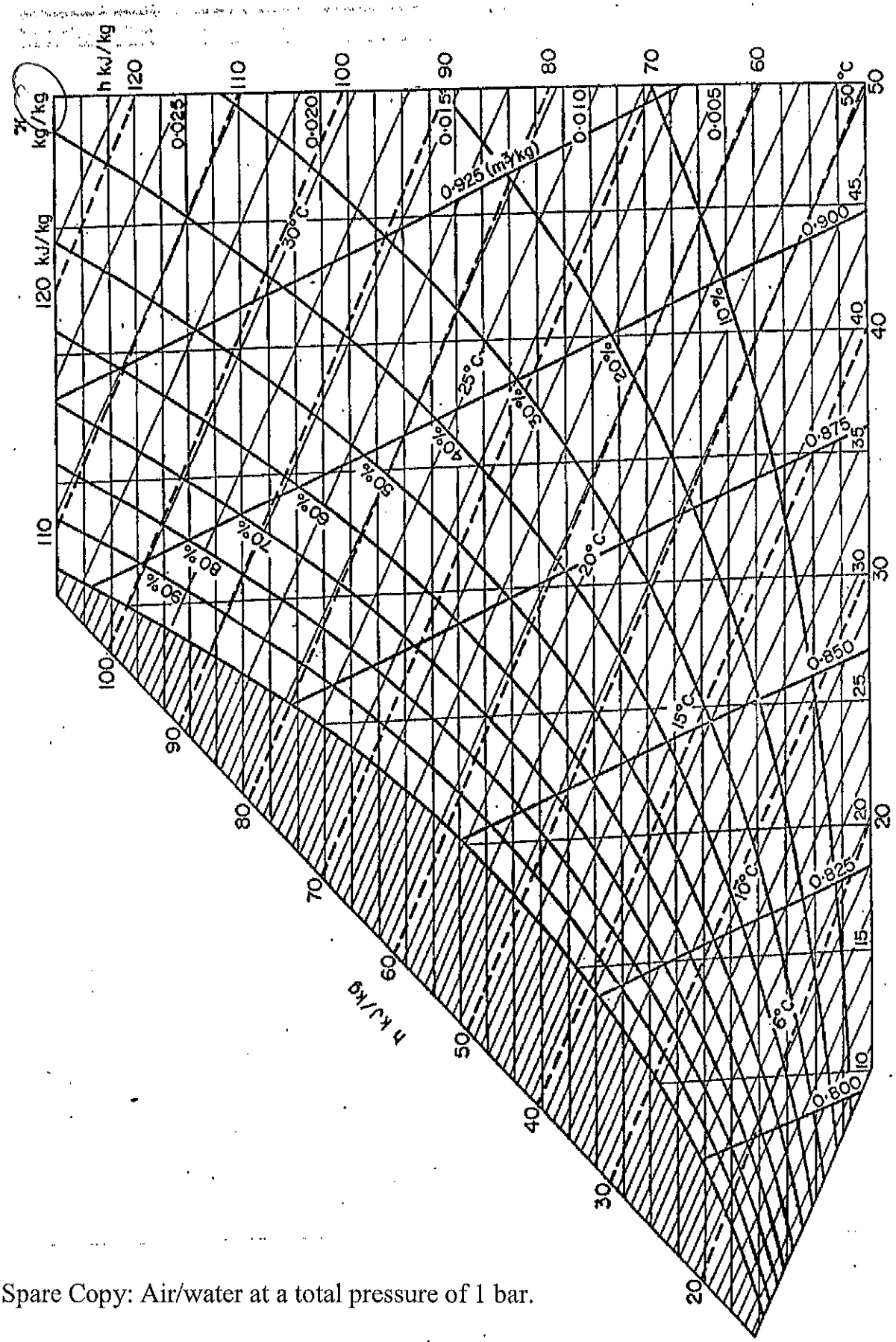
3.2. HUMIDITY CHARTS / PSYCHROMETRIC CHARTS

These charts show thermodynamic data for gas-vapour mixtures, *e.g.* air-water; air-organic.

- The "gas component" (often air) is assumed to be inert and non-condensable,
- the "vapour component" is the gaseous form of a condensable species. Its concentration in the gas phase can vary up to its equilibrium concentration in contact with its (pure) liquid phase p^{SAT}.

A variety of information is summarised on a single humidity diagram (**psychrometric chart**), *e.g.* for air/water at a total pressure of 1 bar:





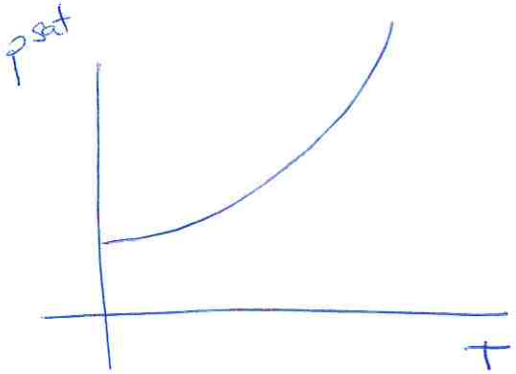
Spare Copy: Air/water at a total pressure of 1 bar.

3.2.1 Humidity

Humidity is the amount of vapour that can be present in a gas phase. It increases with temperature as the partial pressure of the substance increases.

Saturated vapour pressure: the shape of p^{SAT} *vs. T* curves are typically described by the Antoine equation:

$$\ln\left(p^{SAT}\right) = A - \left(\frac{B}{T+C}\right)$$



Rather than describing the amount or concentration of vapour present in terms of a partial pressure, it is more common to introduce the concept of *humidity H*:

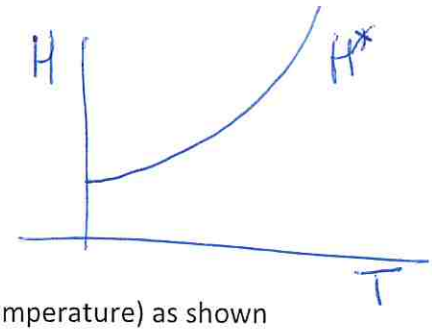
$$H \equiv \frac{\text{mass vapour}}{\text{mass of vapour-free gas}} = \frac{P_v \cdot M_v}{M_w\left(P - P_v\right)} \qquad (2.1)$$

M_v and M_g are the relative molecular mass of the vapour and gas respectively.

"*Vapour-free*" gas is also referred to as "*bone-dry*" gas or "*dry*" gas.

On humidity charts all quantities are normalised to 1 kg of bone-dry gas in the mixture (and not to 1 kg of the mixture itself) *i.e.* they are specified on a *vapour-free basis*.

3.2.2 Saturation Humidity *H**



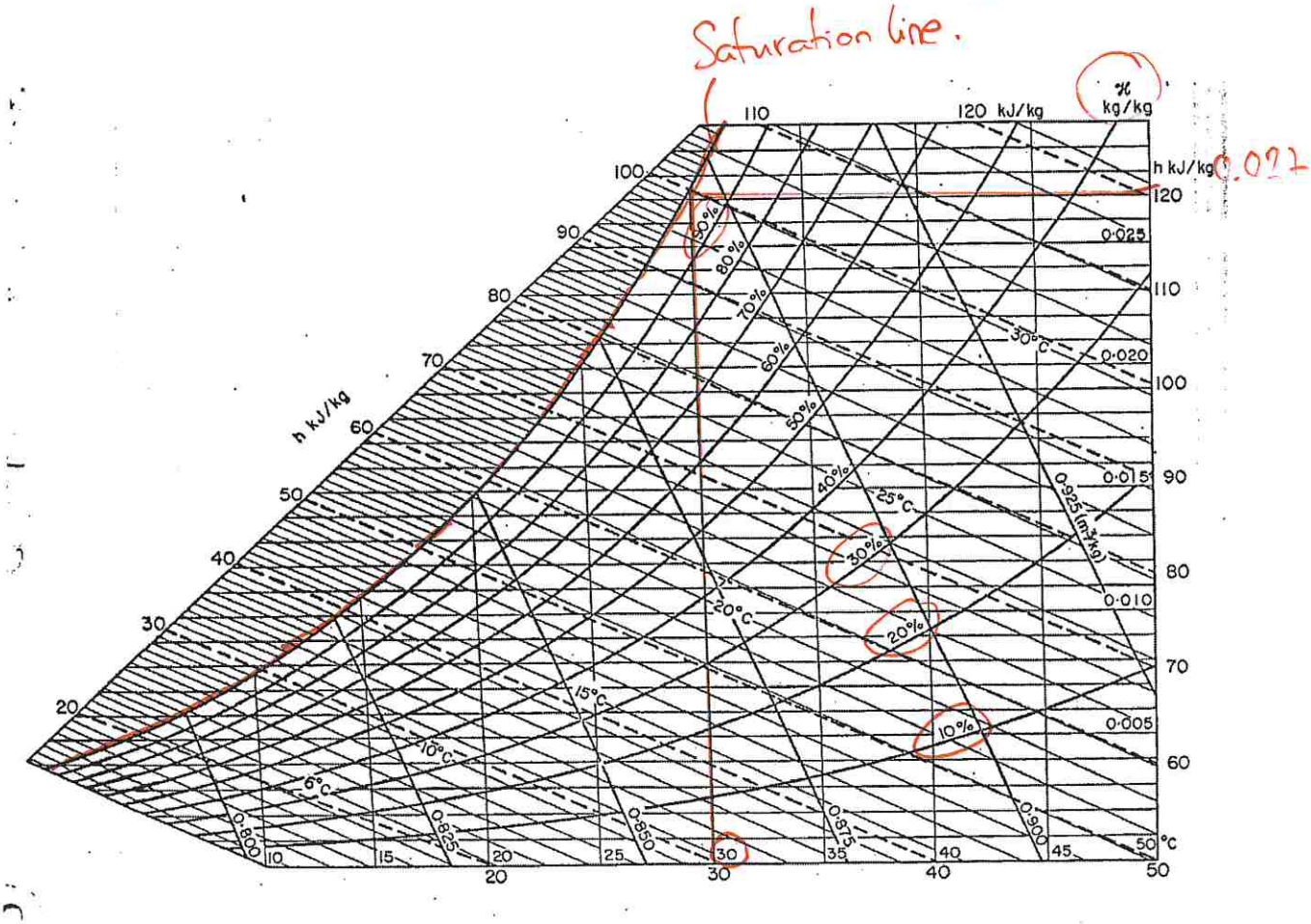
When the mixture of gas and vapour is in equilibrium with the liquid:

$p^v = p^{sat}$ $H = H^{sat}$ or H^*

H^* is a function of temperature (the dry bulb, *i.e.* actual or normal temperature) as shown by the chart.

[Note the nomenclature of using H^* rather than H^{SAT}. Some texts will also use a wiggly H e.g. $\mathscr{H}$ in order to avoid possible confusion with the use of H to denote enthalpy.]

Example: Calculate the saturation humidity at 30°C $H^* = 0.0272$ kg water/Kg dry air



3.2.3 Relative (or percentage) Humidity H_R.

This is the ratio of the actual humidity to the value of H^* at the same T and P:

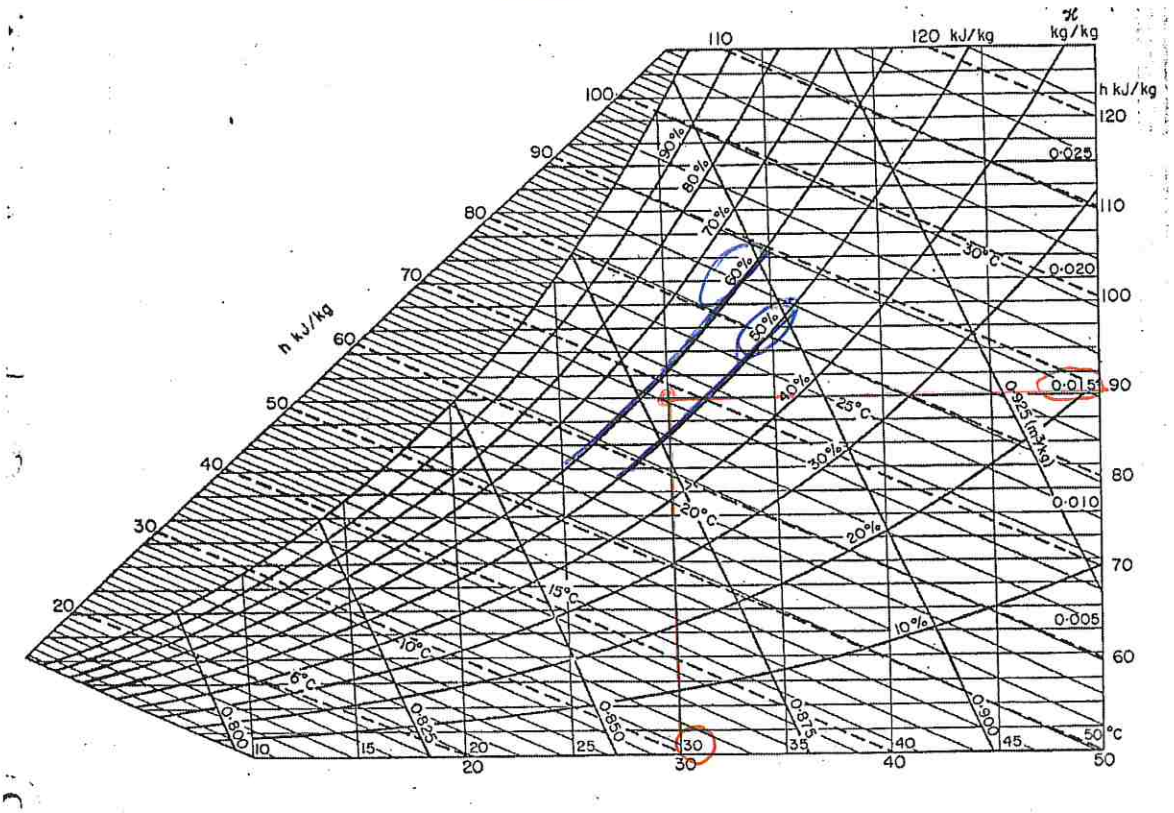
$$H_R = \frac{H}{H^*} \cdot 100\% \qquad (2.2)$$

Relative humidity lines may also be shown on the humidity chart. They have roughly the same shape as the saturation curve and are marked with the % humidity, *e.g.* 90%, 80% *etc.*

Example: Calculate the relative humidity of an air/water mixture at 30 °C and a humidity of 0.015 kg/kg

Solution:

$$H_R = \frac{0.015}{0.0272} \cdot 100 = 56\%$$



3.2.4 Enthalpy of Gas/Vapour Mixture and Humid Heat

The enthalpy *h* of the gas/vapor mixture of humidity *H* is:

h = enthalpy of 1 kg bone dry gas + enthalpy of *H* kg vapour

$$h = C_{p,gas}(T - T_o) + H\left(C_{p,v}(T - T_o) + \lambda\right) \qquad (2.3)$$

where λ is the enthalpy of vaporisation (J/kg) at T_0., a standard state at which the condensable species is in the liquid phase.

This can be re-written as:

$$h = s(T - T_0) + H\lambda \qquad (2.4)$$

where *s* is defined as the *humid heat*, *i.e.* the effective specific heat capacity of this gas/vapour mixture.

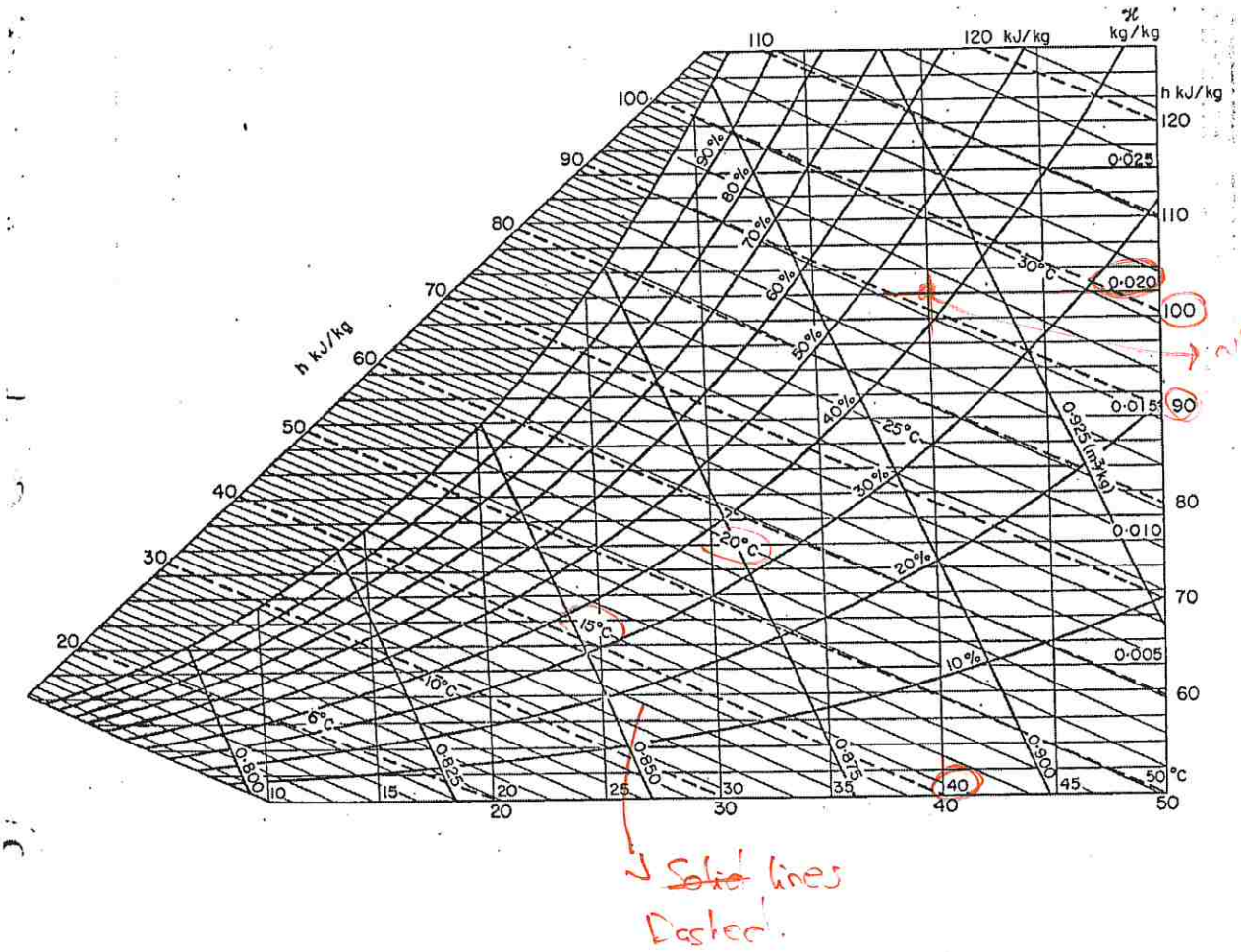
$$s \equiv \left(C_{p,gas} + H \cdot C_{p,vapor}\right) \qquad (2.5)$$

Note: *s* is certainly not a constant as it is a function of *H*.

Enthalpies of gas/vapour mixtures are read from the chart using the "lines of constant enthalpy". These lines are dashed in this diagram and run at an angle (approx.. 45° for air/water) to the principal axes.

Example: Calculate the enthalpy of an air/water mixture at 40 °C and humidity of 0.02 kg/kg.

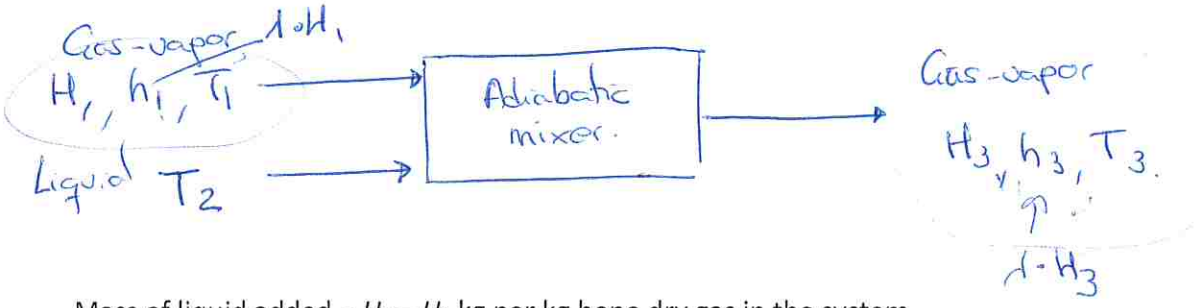
Solution: 92 kJ / kg



Solid lines
Dashed.

Origin of *lines of constant enthalpy* (or *adiabatic saturation lines*).

Consider the mixing of some gas–vapour mixture with some more liquid and allow the system to come to equilibrium in an adiabatic system.



Mass of liquid added = $H_3 - H_1$ kg per kg bone dry gas in the system.

Enthalpy balance (per kg bone dry gas in the system):

$$h_3 = h_1 + (H_3 - H_1)\cdot C_{p,l}(T_2 - T_0)$$

Substitute for h_3 and h_1, using Eqn (2.4):

$$s(T_3 - T_0) + H_3\lambda = s(T_1 - T_0) + H_1\lambda + (H_3 - H_1)C_{p,l}(T_2 - T_0)$$

$$s(T_3 - T_1) = (H_3 - H_1)\{-\lambda + C_{p,l}(T_2 - T_0)\}$$

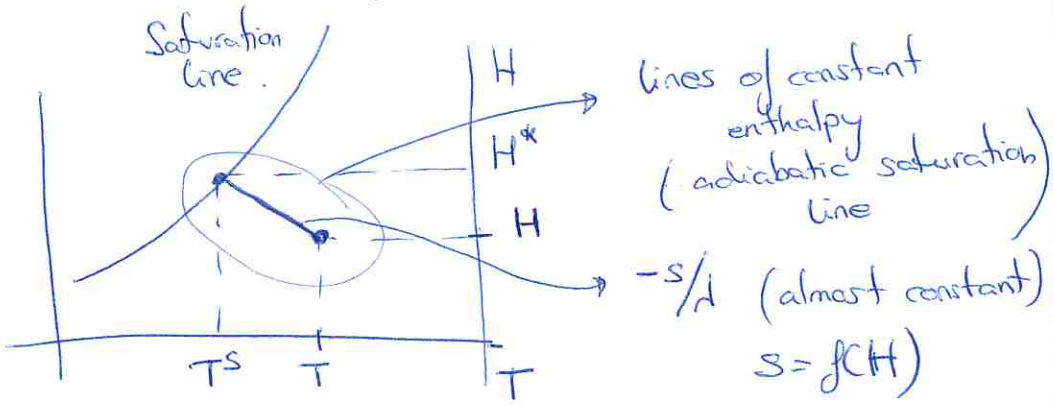
Assume that $C_{p,l}(T_2 - T_0) << \lambda$, (because enthalpy of vaporisation is always much greater than the enthalpy of raising the temperature of the liquid from T_0 to T_2).

$$\frac{H_3 - H_1}{T_3 - T_1} = -\frac{s}{\lambda} \qquad (2.6)$$

This is an *adiabatic saturation line* relating H and T and has slope $-\frac{s}{\lambda}$.

- as more liquid is added, the humidity increases, but the temperature falls as a result of the use of some of the enthalpy in the system to vaporise the added liquid *adiabatically* (i.e. in the absence of an external source of heat),

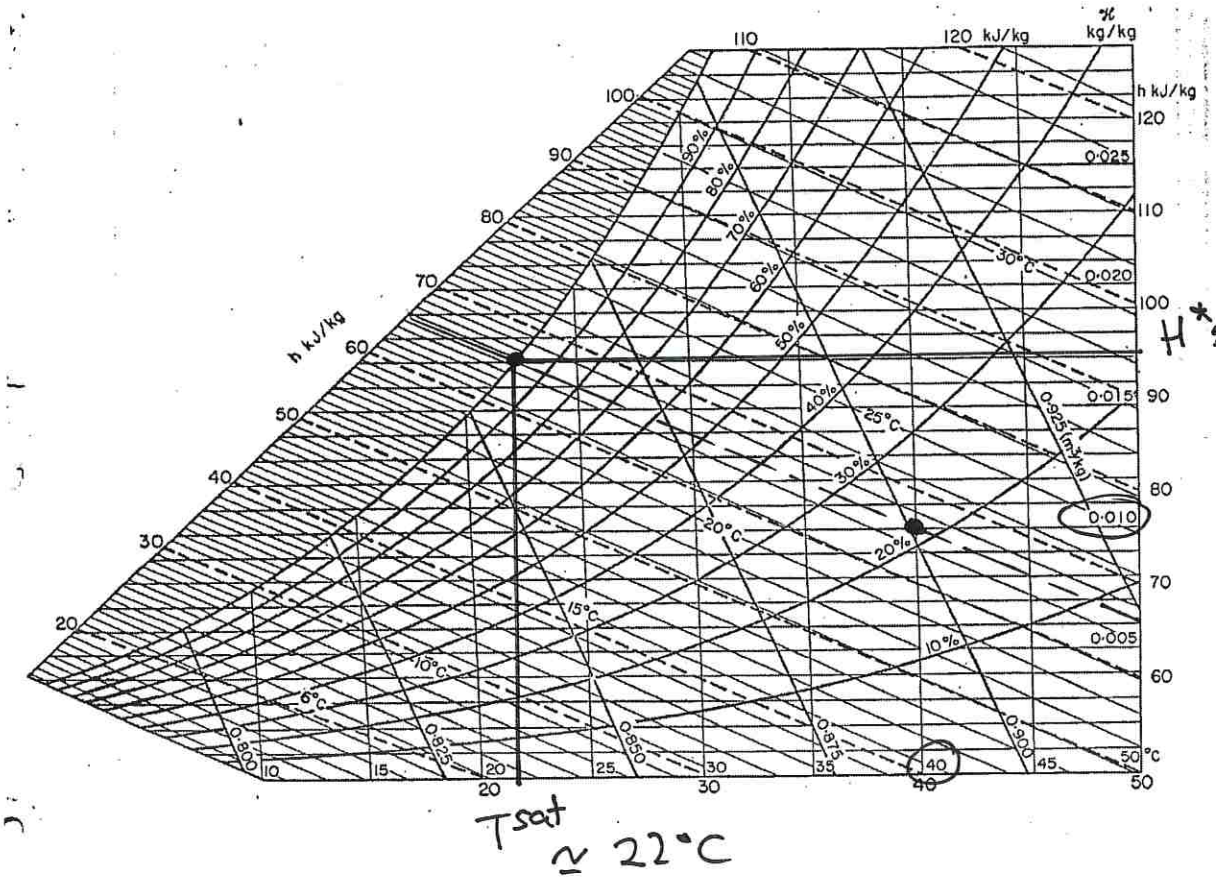
- as more and more liquid is added the system eventually becomes saturated. The temperature and humidity at which this occurs are known as the *adiabatic saturation temperature* T^S and *adiabatic saturation humidity* H^S and can be determined as follows:



Example: Calculate the adiabatic saturation temperature and the adiabatic saturation humidity of an air/water at 40 °C and a humidity of 0.01 kg/kg.

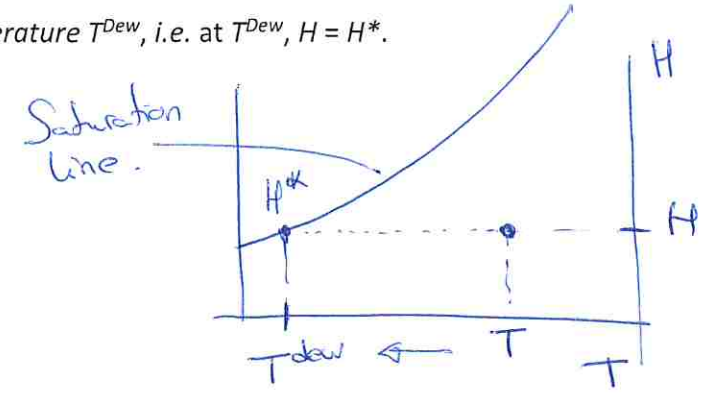
Solution: Adiabatic saturation temperature: 22.5°C

Adiabatic saturation humidity: 0.0174 kg/kg



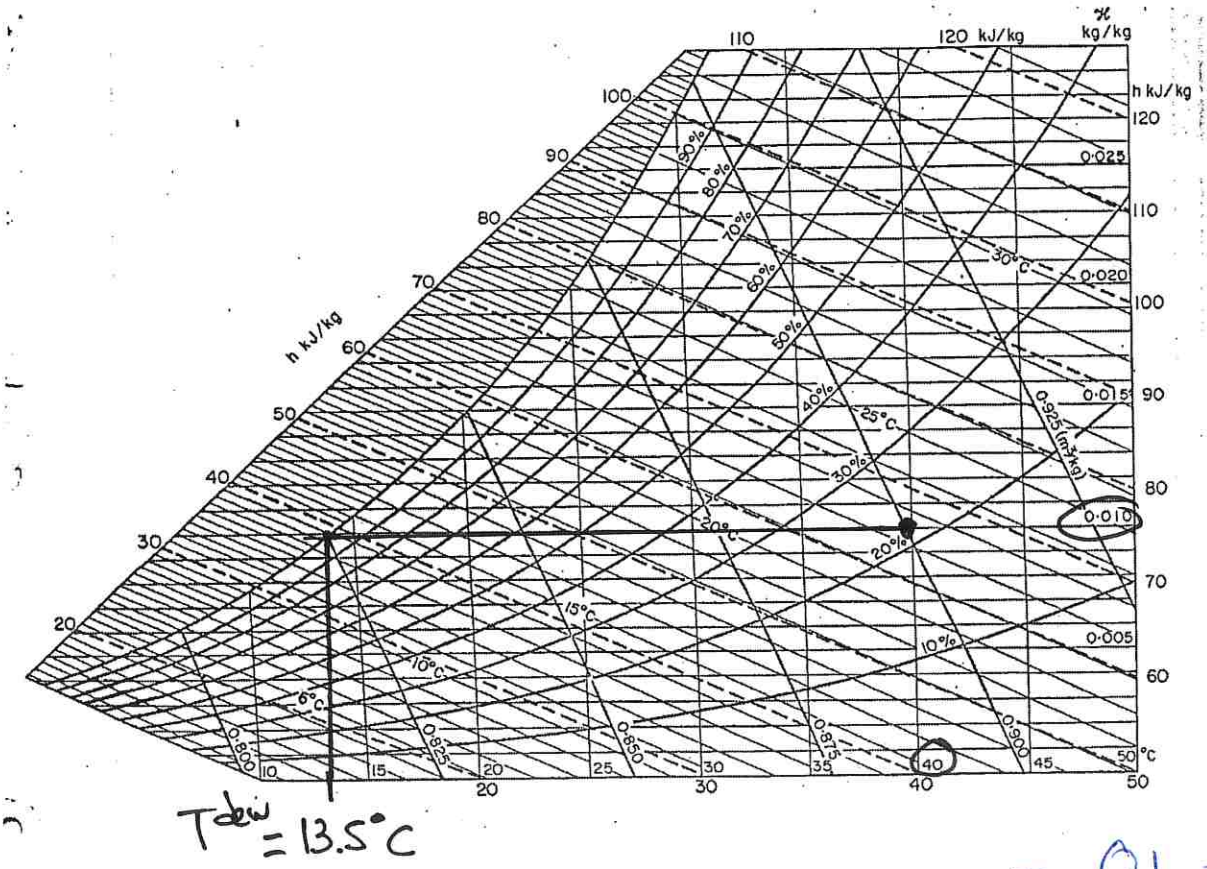
3.2.5 Dew-point Temperature

When a gas/vapour mixture is cooled, the vapour will eventually start to condense and droplets of liquid (dew) appear. The first drop of liquid appears at the *dew point temperature* T^{Dew}, *i.e.* at T^{Dew}, $H = H^*$.



Example: Calculate the dew-point temperature of an air/water mixture at 40 °C and a humidity of 0.01 kg/kg

Solution: Dew-point temperature: ~13.5 °C.



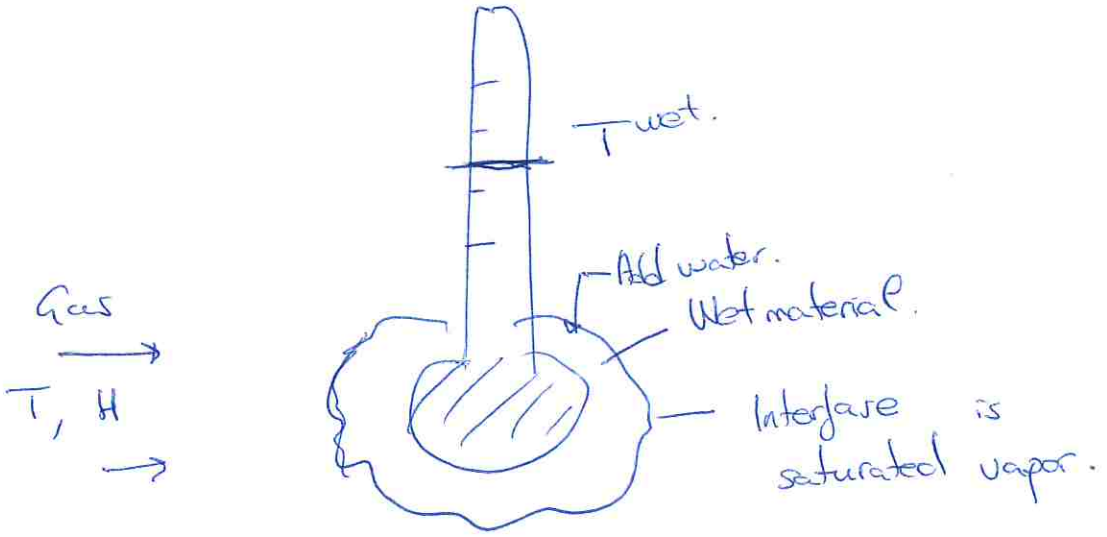
— Q1 —

3.2.6 Wet-bulb Temperature

A wet material exposed to a current of gas takes up a temperature, the *wet bulb temperature* T^W, that is *lower* than the gas temperature, T.

- some of the liquid evaporates from the surface of the material and enthalpy of vaporisation needs to be supplied to drive this process,
- this enthalpy is provided from the surroundings as a result of heat transfer driven by the temperature gradient (T–T^W).

Consider a web-bulb thermometer, *i.e.* an ordinary thermometer with a piece of wet material covering the bulb/sensor.



This is an example of a system in which there is *simultaneous heat* (to supply enthalpy of vaporisation) *and mass transfer* (vaporisation of water).

Rate of *heat* transfer per unit area of cloth:

$$q = U\left(T - T^W\right)$$

Where: U is the overall heat transfer coefficient.

This heat transfer causes some of the liquid to evaporate from the surface of the cloth. Thus, the rate of *mass* transfer per unit area (in mass terms):

$$N_w = M_v k_g \left(c_g^{SAT} - c_g\right)$$

Where: k_g is the mass transfer coefficient (m/s),

c_g is the molar concentration of the vapour in the gas,

c_g^{SAT} is the saturated value at the surface of the material

M_v is the molecular weight of the vapor

Hence using the ideal gas law, $c_g = \frac{n}{V} = \frac{p}{RT}$

$$N_w = k_g \frac{M_v}{RT}\left(p^{SAT} - p\right)$$

Where: p is the partial pressure of the vapour

p^{SAT} is the partial pressure of the vapour at the surface of the material (*i.e.* the saturated vapour pressure at wet bulb temperature T^W).

But from §2.1: $H = \frac{M_v p}{M_g (P - p)} \approx \frac{M_v p}{M_g P}$ for $p << P$, *i.e.* for dilute vapour

and thus: $p = H \frac{M_g P}{M_v}$

i.e. the partial pressure of the vapour can be related to the humidity of the system

$$N_w = k_g \frac{P M_g}{RT}\left(H^W - H\right) = \rho k_g \left(H^W - H\right)$$

or $$N_w = k^{\dagger}\left(H^W - H\right) \qquad (2.7)$$

where: $k^{\dagger} = \rho k_g$ is a mass transfer coefficient (units: kg/m^2 s) appropriate for use with *a humidity difference driving force* ($H^W - H$). ρ = mass density of the gas (kg/m^3).

At steady state:

$$rate\ heat\ supplied = rate\ heat\ used\ in\ vaporisation$$
$$= enthalpy\ of\ vaporisation\ (per\ unit\ mass) \cdot Mass\ transfer\ rate$$

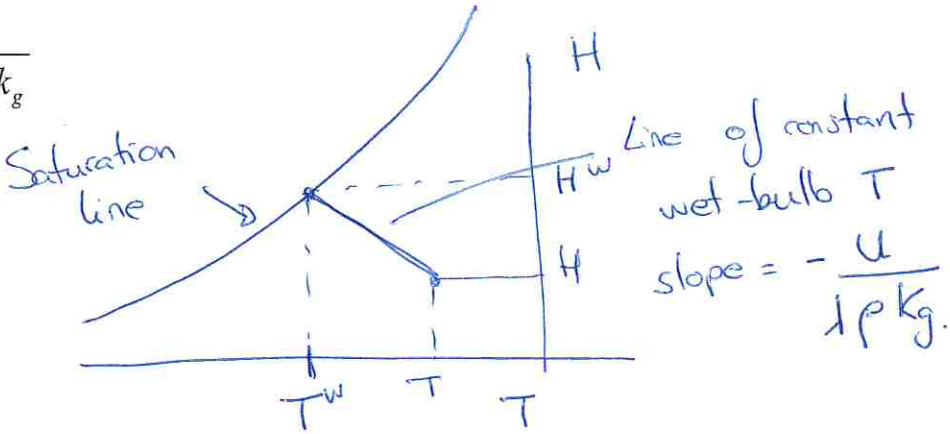
$$q = \lambda N_w$$
$$U\left(T - T^W\right) = \lambda \rho k_g \left(H^W - H\right)$$

$$\frac{H^W - H}{T^W - T} = -\frac{U}{\lambda \rho k_g} \qquad (2.8)$$

slope.

This is the equation of *the line of constant wet bulb temperature* and shows how the *wet-bulb temperature*, T^W, and the *wet-bulb humidity*, H^W, are related to T and H.

The slope of this line is $-\frac{U}{\lambda \rho k_g}$

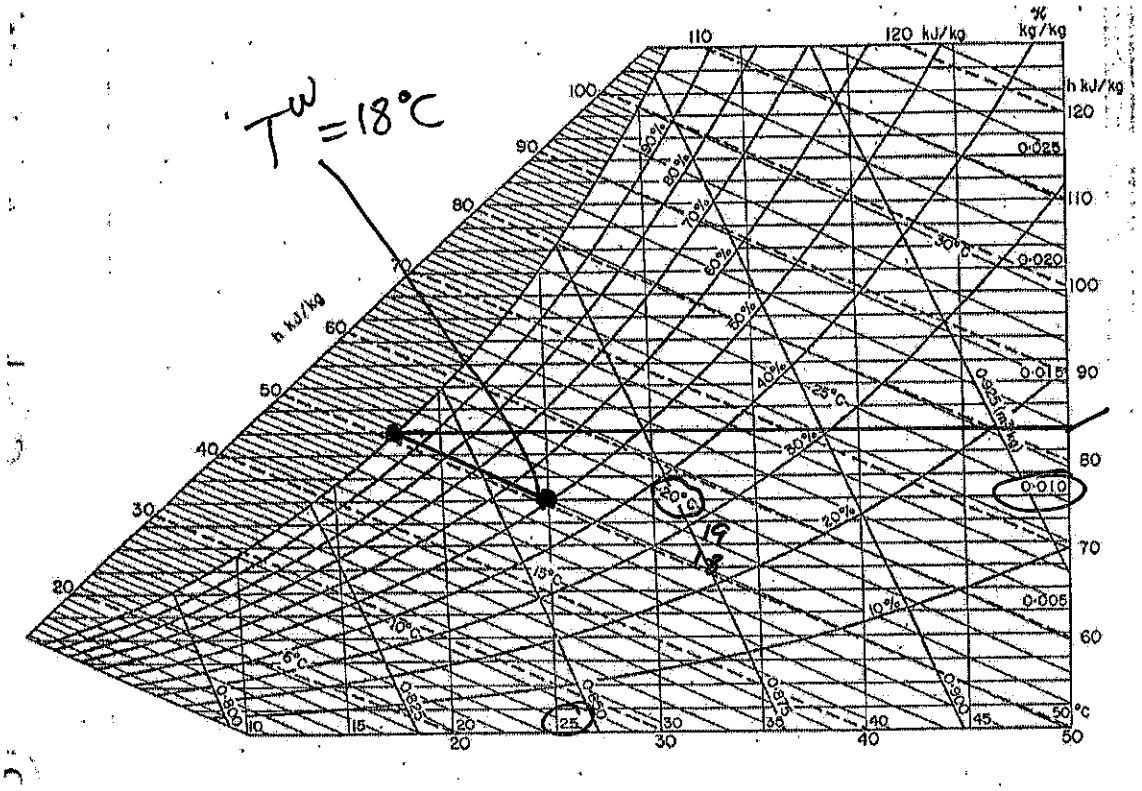


T^W and H^W can be found by going up the appropriate wet-bulb line until it hits the saturation curve. The values of H and T obtained at this intersection are the wet bulb humidity and wet bulb temperatures respectively.

For the air/water system **only**, these lines appear to run parallel to the adiabatic saturation lines (refer to § 2.5, below as to why these lines have approximately the same slope).

Example: Calculate the wet bulb temperature and the wet bulb humidity of an air/water mixture at 25 °C and a humidity of 0.010 kg/kg.

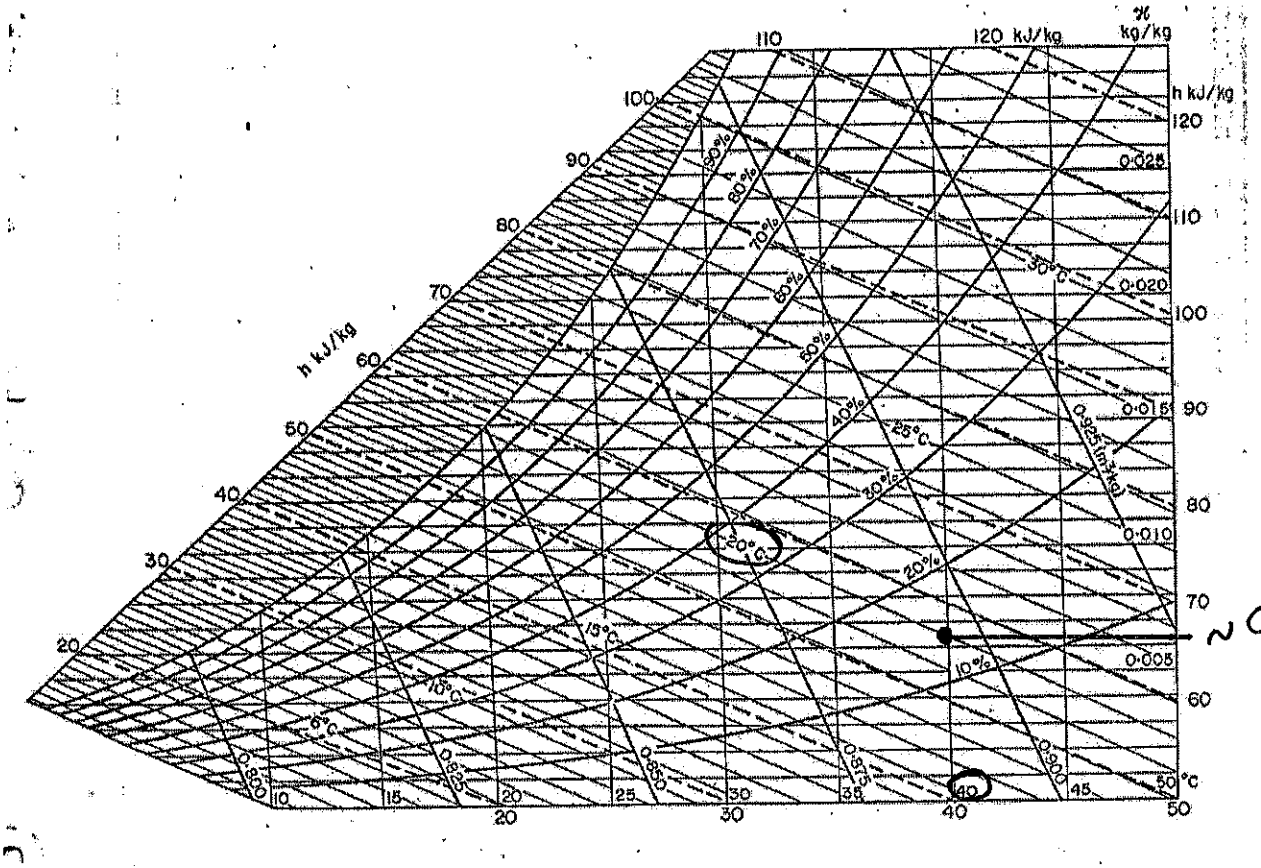
Solution: The wet bulb temperature is 18 °C and the wet bulb humidity is 0.013 kg/kg.



Measurements of the dry bulb (normal) temperature and the wet-bulb temperature enable the humidity of the system to be determined.

Example: Measurements of an air/water system indicated a dry bulb temperature of 40°C and a wet bulb temperature of 20°C; what is the humidity?

Ans.: 0.0064 kg/kg (a relative humidity of ~14%).



≈ 0.0064

3.2.7 Relationship between the Slopes of the Adiabatic Saturation Line and the Wet- Bulb Line

Assume the *j-factor analogy* is applicable, i.e.

$$\mathrm{StPr}^{2/3} = \mathrm{St'Sc}^{2/3} \Rightarrow \frac{\mathrm{St}}{\mathrm{St'}} = \left(\frac{\mathrm{Sc}}{\mathrm{Pr}}\right)^{2/3}$$

Using the definitions (Databook p. 35),

$$\mathrm{St} = \frac{U}{\rho u s} \text{ and } \mathrm{St'} = \frac{k_g}{u} \Rightarrow \frac{U}{\rho k_g} = s\frac{\mathrm{St}}{\mathrm{St'}} = s\left(\frac{\mathrm{Sc}}{\mathrm{Pr}}\right)^{2/3}$$

Then,

$$-\frac{U}{\rho k_g \lambda} = -\frac{s}{\lambda}\left(\frac{\mathrm{Sc}}{\mathrm{Pr}}\right)^{2/3}$$

Slope of the wet-bulb line

Slope of adiabatic saturation line

For an air/water system, $\mathrm{Sc} \approx 1,\ \mathrm{Pr} \approx 1 \Rightarrow \frac{\mathrm{Sc}}{\mathrm{Pr}} \approx 1 \Rightarrow \left(\frac{\mathrm{Sc}}{\mathrm{Pr}}\right)^{2/3} \approx 1$ and hence *for this system only*, the two lines have the same slope as can be seen from the air/water humidity diagram, where the two sets of lines can be seen to be straight and parallel. Under these circumstances:

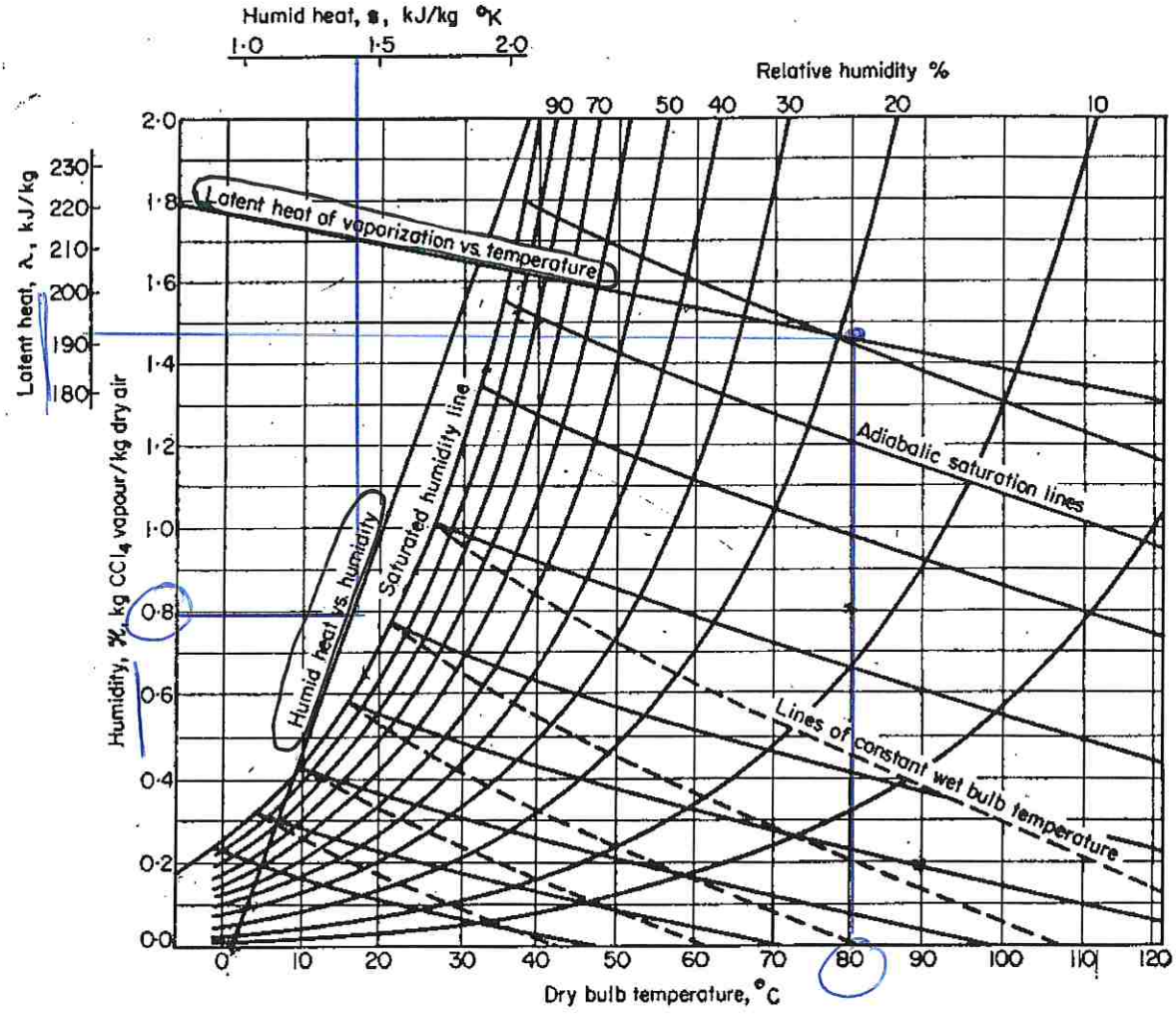
$$U = s\rho k_g$$

humid heat — mass transfer coeff.

heat transfer coeff

i.e. there is a strict relationship between the heat and mass transfer coefficients for air/water mixtures.

For other systems, *e.g.* chloroform/air, the lines have different slopes since $\frac{Sc}{Pr} \neq 1$. Note also that in the latter system the lines are not even straight due to the variations of the physical parameters that comprise the gradients of the two lines as a function of the temperature and humidity of the system.



Humidity chart for chloroform/air

Note: this diagram includes additional data for yet two more useful parameters, namely:

Heat of vaporisation at 80°C. ~ 192 kJ/kg.

i. the variation of the latent heat of vaporisation (λ) with temperature (T)

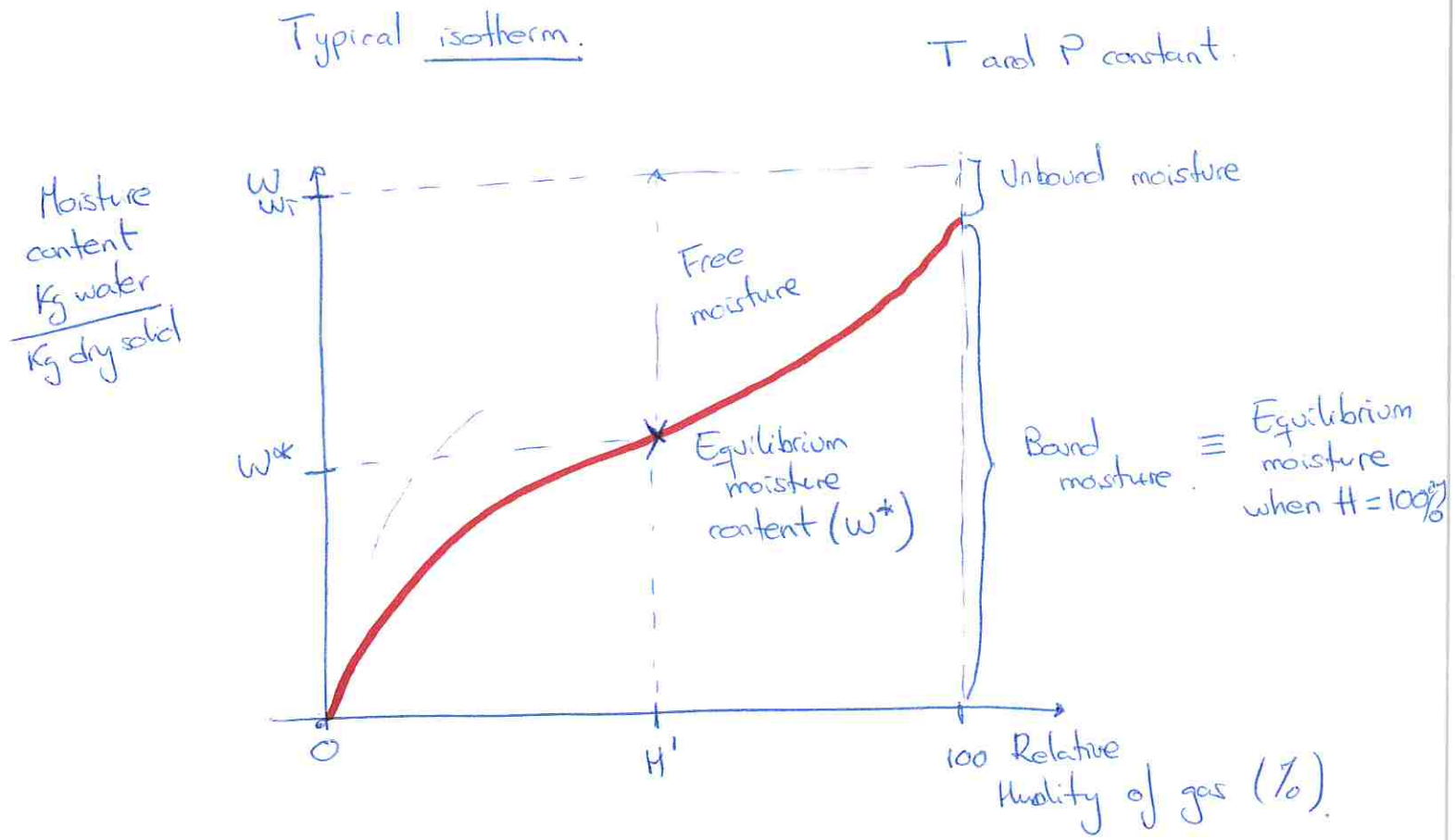
ii. the variation of the humid heat (s) with humidity (H).

Humid heat at H = 0.8 kg/kg. ~ 1.4 kJ/kg K.

Note the appropriate use of the ancillary axes to estimate these quantities.

3.3. MOISTURE CONTENT

The **equilibrium-moisture content** is the amount of water contained by a solid in contact with a gas at a certain temperature, pressure and moisture content of the gas.



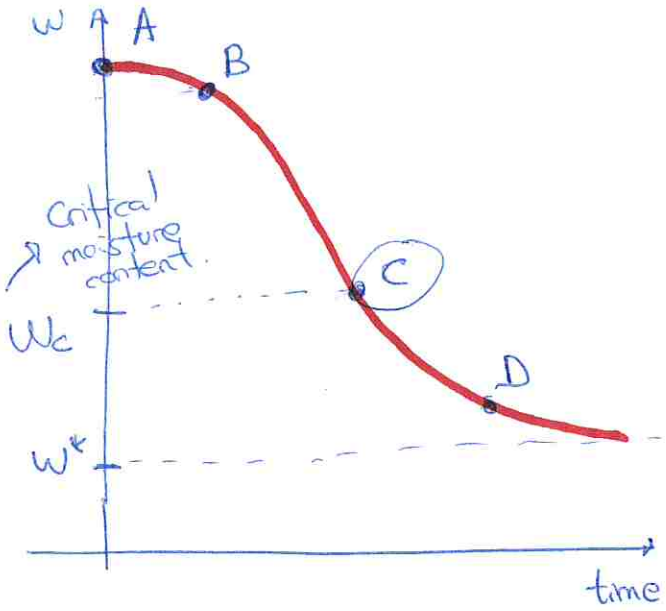
$H = 0 \longrightarrow W \to 0$

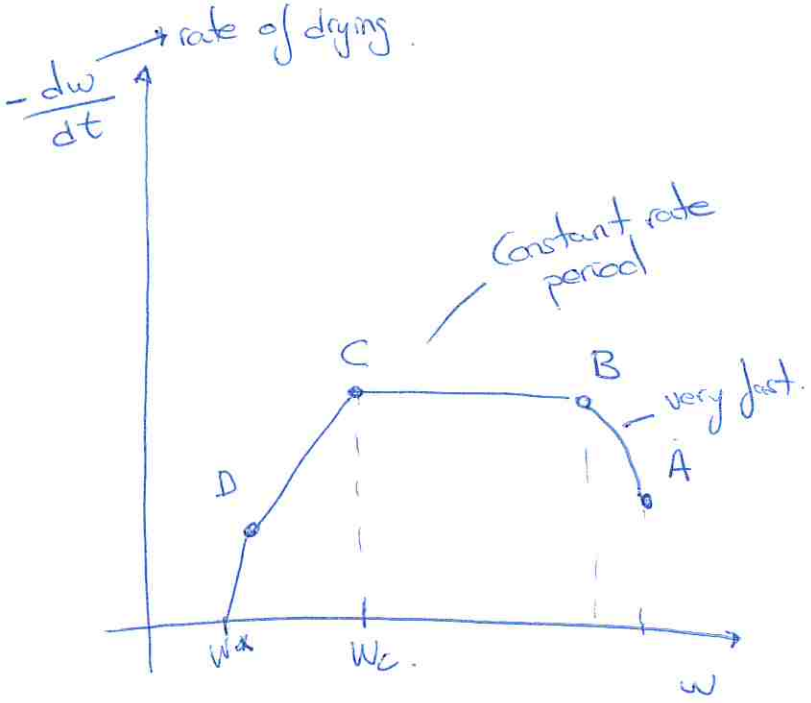
Hygroscopic materials.

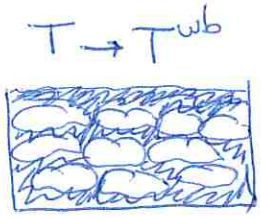
3.4. DRYING PERIODS

The rate at which the liquid can be removed from the material has normally to be determined by experiment.

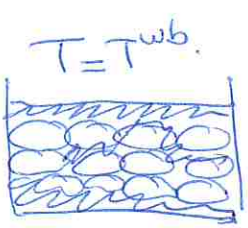
Hence, batch experiments are performed in which a sample of wet material is exposed to a current of gas *at constant temperature and constant humidity* (unlike the situation that occurs in an adiabatic drier, where these parameters change) and the change in the material's liquid content *w* (kg of liquid/kg of bone-dry material) is measured as a function of time, *t*. Note carefully the definition of *w* which appears to be a dimensionless parameter.



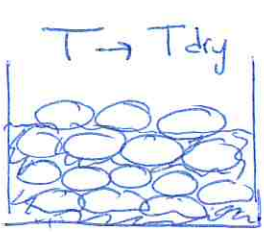




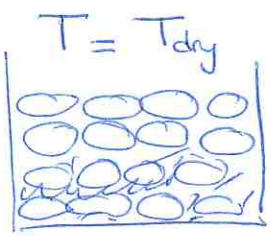
A → B
Initial period
Very fast.



B → C
Constant rate period.



C → D
First falling rate period
Controlled by internal mass transfer.



D → dry (w^*)
Second falling rate period.

One would expect: $-\frac{\mathrm{d}w}{\mathrm{d}t} = f\left(w, H, H^w\right)$, where H^w is the wet-bulb humidity.

The experimental results show two different areas of interest:

(i) The constant rate period

For material with a moisture content > w_c (the so-called *critical moisture content*); the drying rate is independent of w (i.e. it dries at a constant rate).

This regime occurs when the moisture content is sufficiently large for the surface of the material to remain completely wetted; from §2.4, the surface of the material will therefore be at the wet bulb temperature of the gas.

Remember that the wet-bulb temperature T^W will be less than T^G as a result of the need to supply enthalpy to vaporise liquid from the wet surface. By direct analogy with the problem of the wet bulb thermometer (§2.4), the mass flux (kg m^{-2} s^{-1}) will be given by:

$$N_w = k^{\dagger}(H^W - H) \qquad (2.7)$$

where: $k^{\dagger} = \rho\, k_g$ is a mass transfer coefficient (units: kg/m^2 s) appropriate for use with *a humidity difference driving force* ($H^W - H$).

ρ = mass density of the vapour (kg/m^3).

and the total rate of loss of liquid (expressed as per kg of bone dry material) in the constant-rate region will be

$$-\frac{dw}{dt} = k^{\dagger} a_w \left(H^W - H\right) = k^{\dagger} a_w \Delta H$$

where : a_w is the area for mass transfer per kg of bone dry material.

$k^{\dagger}a_w$ can be found either experimentally or sometimes from correlations.

Knowing that in the constant rate period, the rate limiting step is the heat and mass transfer at the solid surface as the solid provides water to the surface at a fast rate, the rate of drying is calculated:

$$(\text{Moisture flux})(\Delta H_{vap}) = \frac{\text{driving force}}{\text{resistance}} = \frac{T - T^{wb}}{1/h}$$

Where h is the heat transfer coefficient

The moisture flux across the surface area of the solid:

$$\text{Moisture flux} = \frac{\text{mass of water removed}}{\text{time} \times \text{Area}} = \frac{1}{A} \cdot \frac{dm_w}{dt}$$

Where m_w is the mass of water removed and A the surface area of the solid

Thus:

$$\frac{1}{A} \cdot \frac{dm_w}{dt} = \frac{h \cdot (T - T^{wb})}{(\Delta H_{vap})}$$

constant.

Integrating:

$$\int_{m_{w,0}}^{m_w,t} dm_w = \int_0^t \frac{h \cdot A \cdot (T - T^{wb})}{(\Delta H_{vap})} \cdot dt$$

$$t_{drying} = \frac{\Delta m_W \cdot (\Delta H_{vap})}{h \cdot A \cdot (T - T^W)}$$

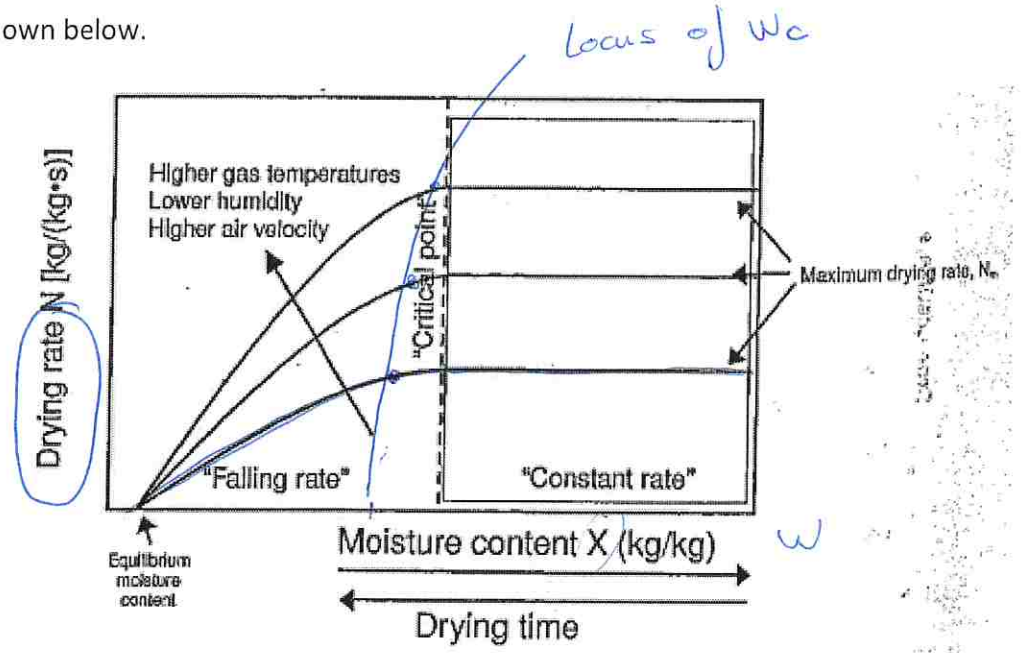
(ii) The falling-rate period $w < w_c$

When the liquid content is less than the critical water content w_c, there is insufficient liquid to wet the surface of the material completely and the material may **not** now be at temperature T^W. $T \neq T^{wb}$

The rate of migration of liquid through the solid to the surface is now the controlling mechanism and the drying rate will now be a function of both w and ΔH and the rate falls as the material dries and w decreases.

Important note. The critical liquid content w_c is not an intrinsic property of the material being dried, and it varies with features such as the thickness of the sample and the rate at which it is being dried.

Typical experimental results from experiments conducted at various imposed values of ΔH are shown below.



In some cases, simplified empirical equations of the following forms can be formulated to describe the drying results with sufficient accuracy.

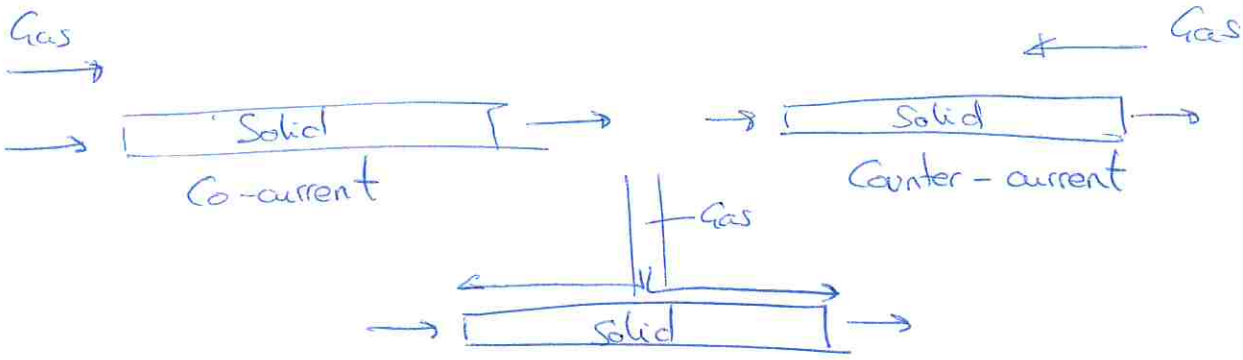
e.g. $-\frac{\mathrm{d}\,w}{\mathrm{d}t} = \beta \Delta H$ during the constant rate period $w > w_c$

$-\frac{\mathrm{d}w}{\mathrm{d}t} = \alpha w$ during the falling rate period $w < w_c$

or $-\frac{\mathrm{d}w}{\mathrm{d}t} = \gamma w \Delta H$ during the falling rate period $w < w_c$

3.5 Calculation of Drying Times in an adiabatic drier

The optimal direction of flow of the gas might be counter-current, co-current or a combination of both.



If the material stays in the constant rate period throughout the drying process, the drying time is the same for all the above flow arrangements, as the driving force is the same at all positions along the length of the drier and is independent of the liquid content of the material.

Example 1: Drying results expressed in analytical form.

A cloth with a drying rate given by:

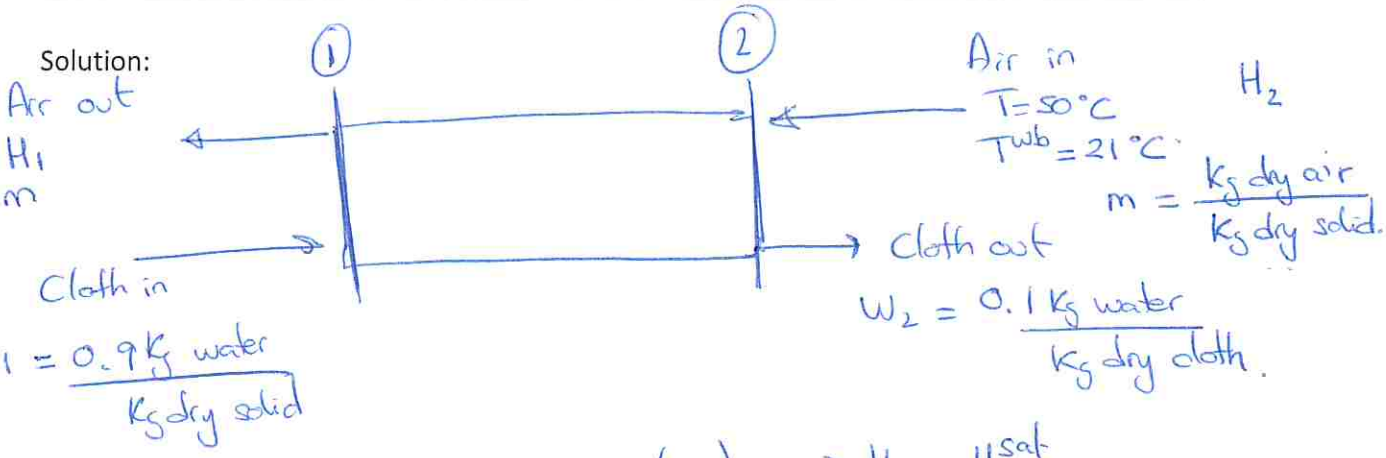
$$-\frac{dw}{dt} = 10\Delta H$$ h^{-1}, in the constant rate period, and

$$-\frac{dw}{dt} = 0.5w$$ h^{-1}, in the falling rate period

is to be dried from a water content of 0.9 kg water/kg bone dry cloth to w = 0.1 kg/kg in a counter-current, adiabatic drier. Air is provided at 50 °C with a wet bulb of 21 °C.

a. Calculate the drying time if the air flowrate is 1.2 × minimum.
b. Calculate the humidity of the exiting air

Solution:



i. Overall mass balance of water

$(m)_{min} \Rightarrow H_1 = H^{sat}$

Let air flowrate be *m* kg bone dry air/kg bone dry cloth (a flow rate that will (conveniently) remain constant all along the length of the drier).

Basis: 1 Kg solid

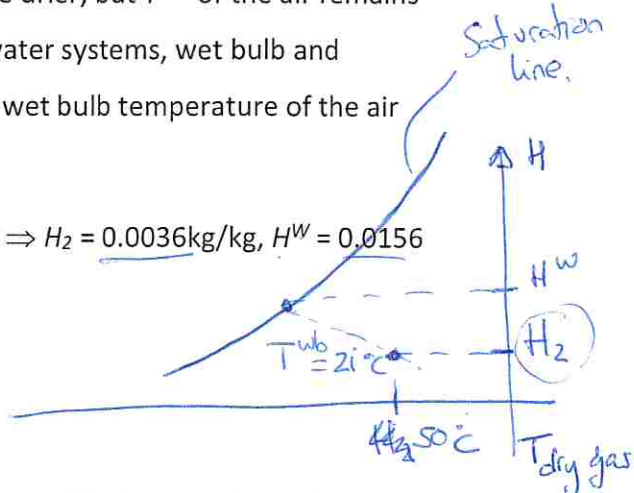
1. $$w_1 + m\,H_2 = w_2 + m\,H_1$$

$$m(H_1 - H_2) = w_1 - w_2$$

$= H^{sat}$

ii. From the humidity chart, calculate the saturation humidity of the air (H^W) and H_2

The humidity *H* of the air increases as it moves through the drier, but T^{SAT} of the air remains constant (the adiabatic saturation temperature). For air/water systems, wet bulb and adiabatic saturation temperature lines overlap, hence the wet bulb temperature of the air also remains at its inlet value. T^W = 21 °C.



Find H_2 and H^W from humidity chart: T_2 = 50 °C, T^W = 21 °C ⇒ H_2 = 0.0036kg/kg, H^W = 0.0156 kg/kg.

iii. Calculation of the minimum air flow (m_{min})

For minimum air flowrate, m_{min}, the outlet air will just be at equilibrium with incoming material, *i.e.* this air will be saturated ⇒ $H_1 = H^W$ = 0.0156 kg/kg.

So:

$$m_{min} = \frac{(w_1 - w_2)}{H^W - H_2} = \frac{0.9 - 0.1}{0.0156 - 0.0036} = 66.7\ \frac{kg\ dry\ air}{kg\ dry\ cloth}$$

Actual air rate = 1.2 × 66.7 = 80 kg/kg.

iv. Mass balance around control section:

$$m(H - H_2) = w - w_2$$

and since H^W is constant (because this is an air-water system):

$$m([H - H^W] - [H_2 - H^W]) = w - w_2$$

where $\Delta H = H^W - H$ driving force.

$$m(\Delta H_2 - \Delta H) = w - w_2$$

$\Delta H_2 = H^W - H_2$ = 0.0156 - 0.0036 = 0.012 $kg_{water}/kg_{dry\text{-}air}$

Hence:

$$\Delta H = \Delta H_2 - \frac{1}{m}(w - w_2) = 0.012 - \frac{1}{80}(w - 0.1)$$

$$\Delta H = 0.01325 - 0.0125w \quad \text{Operating line.} \qquad (1)$$

This equation is an *operating line* relating values of ΔH and w at any point in the drier.

v. Find critical water content (w_c)

The critical water content that occurs at the intersection of the constant rate and falling rate periods, i.e. the point where the drying rate is the same in both regimes:

$$-\frac{dw}{dt} = 10\Delta H = 0.5w_C \qquad (2)$$

Combine eqs. (1) and (2) to find, w_C = 0.212 kg water/kg dry cloth.

vi. Drying time during constant rate period

The total drying time is the sum of the times spent in the constant rate and falling rate periods. During the constant rate period:

$$-\frac{dw}{dt} = 10\Delta H = 0.1325 - 0.125\,w \qquad (3)$$

so the drying time in the "constant rate period" i.e. from w_1 to w_C:

$$t_{const} = \int_{w_1}^{w_c} \frac{dt}{dw}\cdot dw = \int_{0.9}^{0.212} \frac{dw}{0.125\,w - 0.1325} \qquad = 13.3 \text{ h}$$

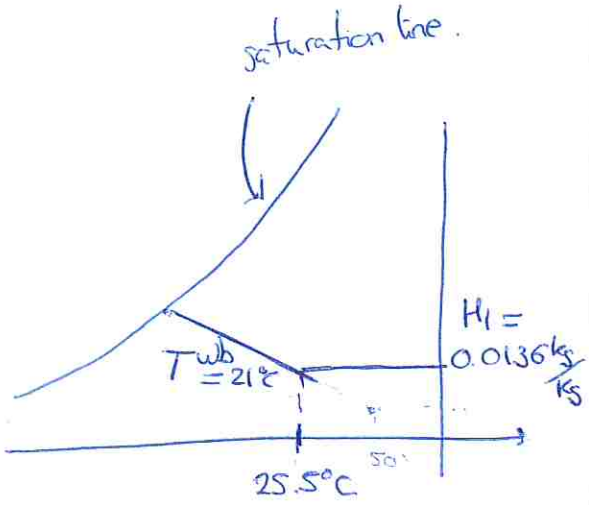
vii. Drying time during the falling rate period

$$-\frac{dw}{dt} = 0.5w$$

hence the drying time in the falling rate period i.e. from w_C to w_2

$$t_{fall} = \int_{w_c}^{w_2} \frac{dt}{dw}\cdot dw = \int_{0.212}^{0.1} \frac{-dw}{0.5w} \qquad = 1.5 \text{ h.}$$

Hence the total drying time is 14.8 h.



viii. Calculate the actual humidity of the exiting air

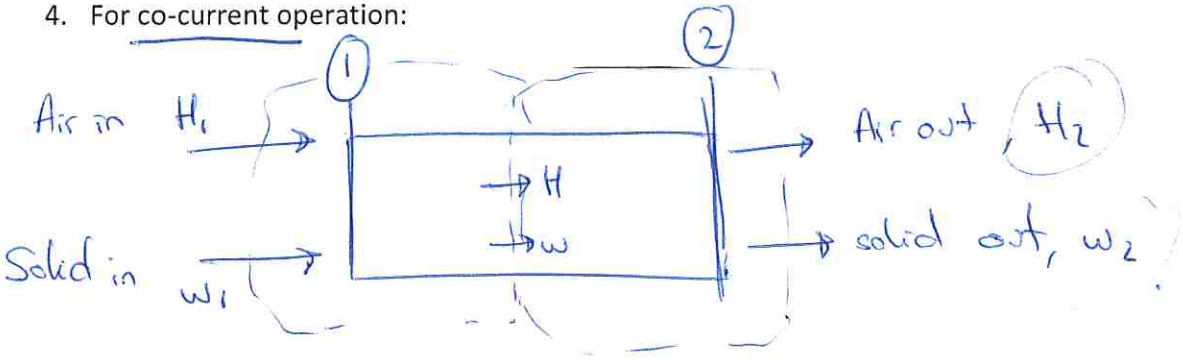
$$m\,(H_1 - H_2) = w_1 - w_2$$

$$H_1 = \frac{0.9 - 0.1}{80} + 0.0036 = 0.0136 \frac{kg\ water}{kg\ dry\ air}$$

Notes:

1. The dry bulb temperature of the air at any point in the drier can be read off the adiabatic saturation line at the appropriate value of *H*. *e.g.* the exit air (H_1 = 0.0136) is at 25.5 °C.

2. In a continuous drier, the rate of drying is not actually constant whilst the cloth is in the so-called constant rate period. In this period the rate is proportional to ΔH and ΔH varies along the drier (remember H^W remains constant along the drier for an air/water system).
3. To find the final liquid content within the cloth knowing the total drying time needs a trial and error, iterative solution, since the limits on the integrals for the two drying periods are not known (because w_C is not a constant and cannot be predicted).
4. For co-current operation:

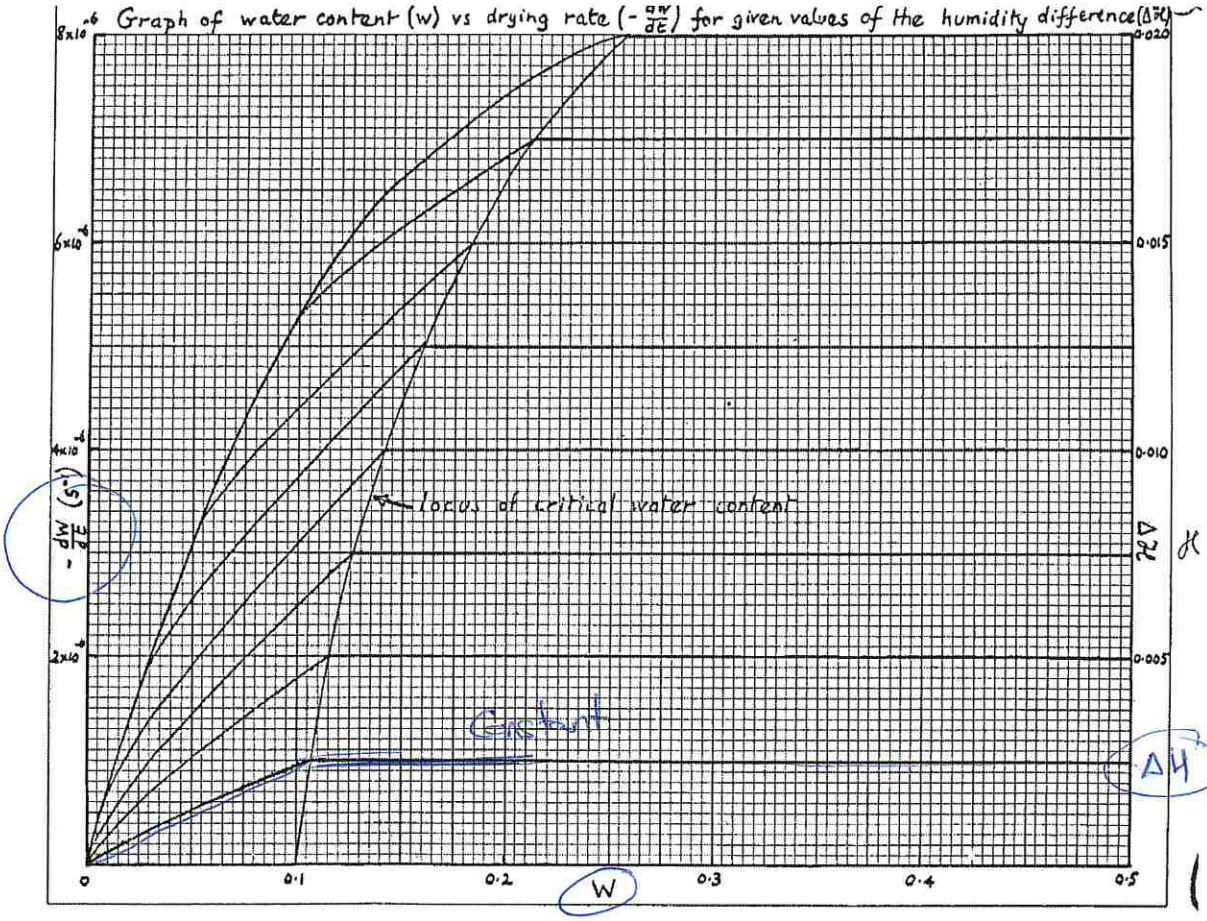


Operating line is now:

$$\Delta H = \Delta H_2 + \frac{1}{m}(w - w_2)$$

otherwise, the procedures are the same.

Example 2: Drying results expressed in graphical form



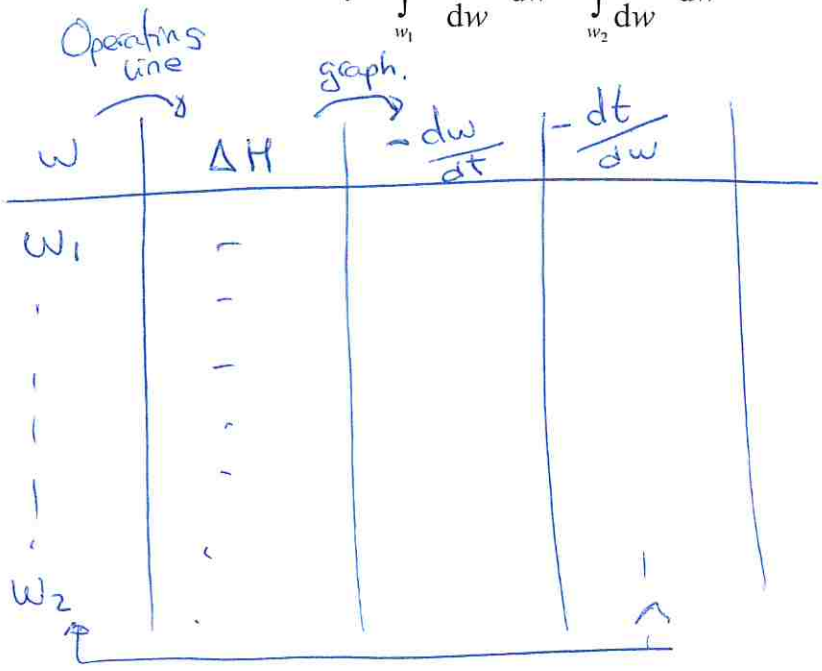
From Example 1, for a counter-current drier the operating line is:

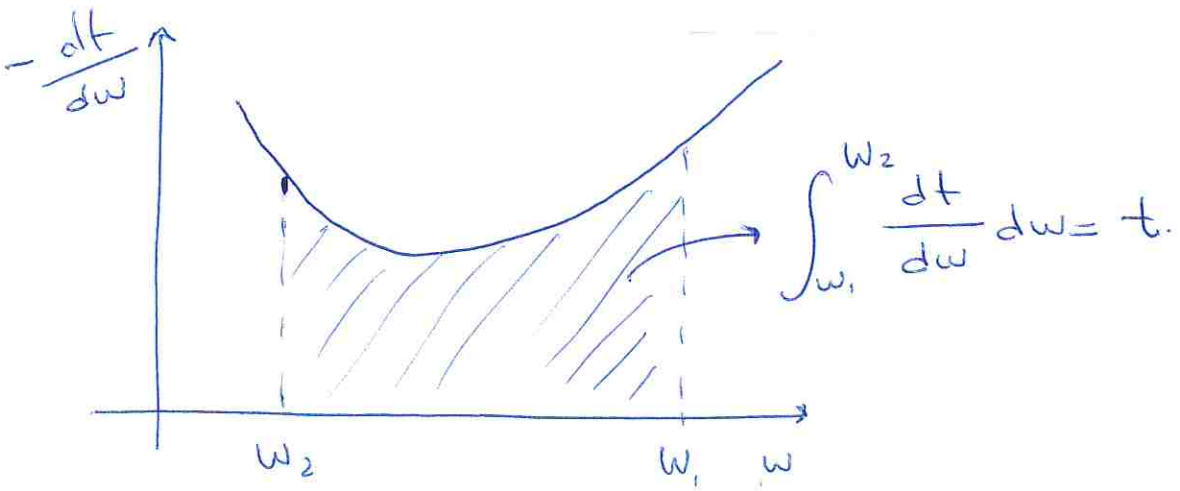
$$\Delta H = \Delta H_2 - \frac{1}{m}(w - w_2)$$

- For each of the values of ΔH pertaining to a line in the chart (in this case, 0.0025, 0.005, 0.0075, 0.010, 0.0125, 0.015, 0.0175, 0.020), calculate the associated value of w using the operating line equation.
- For each value of w so obtained, use the chart with the associated ΔH line, to read off the value of $-\frac{dw}{dt}$ for that point in the drier:

- Plot the reciprocal $-\frac{dt}{dw}$ *vs. w* and integrate graphically to find the total drying time.

$$t = \int_{w_1}^{w_2} -\frac{dt}{dw} \cdot dw = \int_{w_2}^{w_1} \frac{dt}{dw} \cdot dw$$

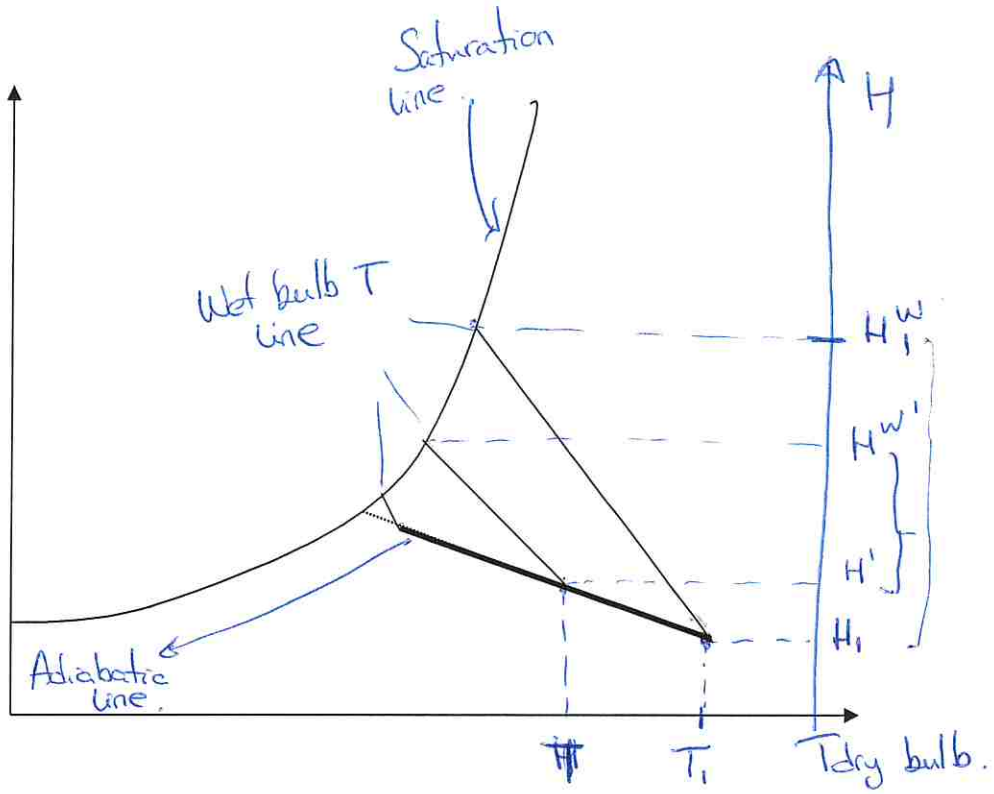




3.6 Drying Times for Solvents Other Than Water

In adiabatic cross-flow driers, the gas continues to follow a line of constant enthalpy as it moves through the drier.

However, as the slopes of the constant enthalpy line and the constant wet bulb line are different, the wet bulb temperature of the gas falls along the length of the drier as the gas becomes increasingly saturated with evaporated vapour. Hence, H^W is no longer constant and ΔH has to be evaluated at various positions along the drier.



We cannot use the ΔH vs. w form of the operating line under these circumstances (as H^W is no longer constant).

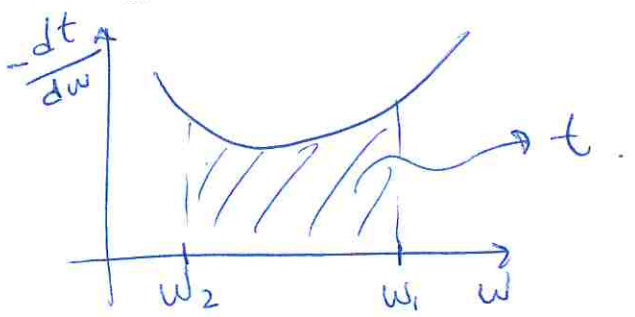
But e.g. for a counter-current systems, we can still use an operating line of the form:

$$m(H - H_2) = w - w_2$$

Hence we prepare a table containing values of the following variables between the inlet and outlet conditions

w	H	H^W	ΔH	$-\frac{dt}{dw}$
w_1	H_1	$H^W{}_1$	$H^W{}_1 - H_1$	$-\frac{dt}{dw}$
.	.	.	.	.
.	.	.	.	.
w	H	H^W	$H^W - H$	$-\frac{dt}{dw}$
.	.	.	.	.
.	.	.		..
w_2	H_2	$H^W{}_2$	$H^W{}_2 - H_2$	$-\frac{dt}{dw}$

- The values of H are obtained from the operating line for the appropriate value of w.
- The values of H^W are obtained from the humidity chart using the fact that the associated H values will lie on the adiabatic line that runs through the gas conditions at the entry to the drier.
- The values of $-\frac{dt}{dw}$ are obtained from the drying chart (or empirical equations) at the appropriate value of ΔH and w.
- As before, plot the reciprocal $-\frac{dt}{dw}$ *vs.* w and integrate graphically to find the total drying time.



3.7. Non-Adiabatic Driers

In these cases, the calculations become more difficult and tedious as the following aspects should be taken into consideration:

- The material is no longer at the wet bulb temperature of the gas
- The material receives heat from other sources than warm gas
- At any point in the drier, the temperature of the material, and the associated saturation humidity, have to be found from an enthalpy balance.